

Distortion of Perceived Visual Space after Prolonged Horizontal Eccentric Gaze Holding

Terence L. Tyson^{1,2,3}, Dennis F. Perez¹, and Jorge Otero-Millan^{1,2,4}

¹Herbert Wertheim School of Optometry and Vision Science, University of California, Berkeley, Berkeley, CA; ²Department of Neuroscience, College of Letters and Science, University of California, Berkeley, Berkeley, CA; ³Human Systems Integration Division, NASA Ames Research Center, Moffett Field, CA; ⁴Department of Neurology, Johns Hopkins University, Baltimore, MD

Correspondence to:

Mr. Terence L. Tyson

Herbert Wertheim School of Optometry and Vision Science

University of California, Berkeley

Berkeley, CA

terencetyson@berkeley.edu

Key words: Gaze holding, visual space, eye position, adaptation

Abstract

Eye movements have long been used as a measure of underlying brain function and pathology. A specific kind of eye movement, named rebound nystagmus (RN), has been shown to be a behavioral window into adaptation mechanisms of ocular motor control. Rebound nystagmus is an eye movement aftereffect of prolonged eccentric gaze holding. Upon returning to central fixation the eyes drift or “rebound” back toward the previously-held gaze location, demonstrating an adaptive process. Little is known about how gaze holding and its consequential adaptation of the oculomotor system affects the perception of visual space. Here, we used a variant of the landmark task to assess spatial bias (or lack thereof) with and without eccentric gaze holding. We found that perceived spatial bias after prolonged eccentric gaze holding was significantly different between gaze holding to the far left (-40 deg) and the far right (+40 deg). We also found that sensitivity in distinguishing relative distances between objects in space was marginally different between the left and right gaze holding conditions. This suggests that perceived visual space is differentially impacted by where gaze was previously held, reflecting a dependence on the history of eye positions.

1 Introduction

The gaze-holding system enables stable gaze and in turn a stable percept of reality. However, imperfections exist in this circuit, with studies suggesting that these imperfections are a feature and not a bug. For example, the random-walk drift during gaze fixation has been shown to enhance dynamic visual acuity (Anderson et al., 2020; Clark et al., 2022) and improve reconstruction of visual percepts in the brain (Anderson et al., 2020). Another salient imperfection has been shown in gaze holding at eccentric eye positions where the drift motion is not random in direction, but tends toward central fixation (0 deg). This particular eye movement, commonly referred to as gaze-evoked nystagmus (GEN), exhibits a periodic pattern of slow drift toward the center followed by a saccade in the opposite direction to correct for the drift.

This ocular motor phenomenon is typically interpreted as an imperfection of the gaze-holding system. Specifically, this directional drift arises from the inherent leakiness of the oculomotor integrator in which the accumulated neural drive for eye position dissipates over time (Leech et al., 1977; Leigh & Zee, 2015; Otero-Millan et al., 2019). This leakiness property can prove useful in conferring resiliency from perturbation in the presence of noise, since the ideal integrator should avoid accumulating noise while retaining the relevant signal.

Gaze-evoked nystagmus is often analyzed by measuring how the drift (or slow-phase) velocity changes as a function of eye position (Bertolini et al., 2013, 2019). The inverse slope of this position-velocity curve corresponds with the estimated *time constant* of the leaky integrator.

Becker and Klein showed a time constant of 25 seconds for horizontal displacements of eye position in humans (Becker & Klein, 1973; for review see Sanchez & Rowe, 2018), reflecting a slow decay back to central fixation.

The performance of the gaze-holding integrator is not constant and can adapt over time. First, the centripetal drift becomes smaller over time during prolonged eccentric gaze holding, suggesting that the gaze-holding system is *adapting* to the current eye position (Zee et al., 2017). Second, evidence of adaptation in the integrator has been shown in the manipulation of the gain or phase of the vestibular-ocular reflex (Kramer et al., 1995; Tiliket et al., 1994), and retinal-slip feedback (Arnold & Robinson, 1991, 1997; Major, Baker, Aksay, Mensh, et al., 2004; Major, Baker, Aksay, Seung, et al., 2004; Turaga et al., 2006). Finally, rebound nystagmus has been used as a behavioral window into the *adaptive* properties of the oculomotor integrator by studying the aftereffect that occurs after holding prolonged fixation at an eccentric point and upon return to central gaze the eyes would drift toward the previously held position (Leigh & Zee, 2015; Otero-Millan et al., 2019; Shallo-Hoffmann et al., 1990). Oculomotor behavior has proven to be an insightful measure of adaptive processes in the oculomotor system, specifically of the neural integrator. However, how does perception of space change with this adaptive process? Does it change with the integrator, and if so, does it change on a similar timescale? One approach to probing changes in the perceived perception of space would be to observe visuospatial asymmetries or spatial biases after holding eccentric gaze to a particular side. There exists a large corpus of findings on spatial bias perception in normal and patient populations using various paradigms (Learmonth & Papadatou-Pastou, 2022), the most

common being Milner's landmark task (Milner et al., 1993) and Schenkenberg's bisecting line task (Schenkenberg et al., 1980). These approaches have mainly shown a consistent leftward spatial bias where space on the left side of central gaze appears to be larger than it actually is (Jewell & McCourt, 2000); this perceptual asymmetry is often termed as *pseudoneglect*, signifying that the right hemifield of space is perceptually neglected while the left hemifield garners more perceptual resources (Bowers & Heilman, 1980).

Here we set out to better understand the link between the effects of holding eccentric gaze on the perception of visual space, and more specifically, the connection between oculomotor adaptation and the perceptual adaptation of perceived visual space. We probed this potential link using a variant of Milner's landmark task, where the stimulus is already "pre-bisected" and the observer has to determine whether the midline is closer to the left or right line. Observers completed the task while looking centrally (0 deg eccentricity) and *after* holding their gaze eccentrically either to the left or right at 40 degrees eccentricity and then performing the judgment upon returning to central gaze. How does the perception of space change after the oculomotor integrator adapts to a new eye position? Here, we studied whether perception of space changes after the oculomotor integrator adapts to a new eye position.

2 Methods

2.1 Subjects

Twelve subjects completed the study and were compensated for their participation. All subjects completed a consent form that was approved by the institutional review board (IRB) at the University of California, Berkeley and followed the guidelines established by the Declaration of Helsinki. Visual acuity, ocular dominance, and hand dominance were assessed prior to the main task. All subjects had normal or corrected-to-normal binocular visual acuity better than 20/40 Snellen fraction or 0.3 logMAR and were assessed using a Snellen eye chart (Snellen, 1862). For a comprehensive list of subject demographics, see Table 1.

Subject Demographics	
<i>n</i>	12 (6 females)
Age (years)	30.1 ± 3.5
Handedness Score*	81.3 ± 43.1
Dominant Eye†	7 OD and 5 OS

*Handedness categories from a short-form version of the Edinburgh Handedness Inventory (Oldfield, 1971; Veale, 2014): right handers = 61 to 100, mixed handers = -60 to 60, left handers = -100 to -61

†Miles test (Miles, 1929)

115 2.2 Equipment

116 For the main experiment, subjects were asked to fix their head within a forehead and chin rest
117 while viewing visual stimuli presented on a display monitor at an 83 cm viewing distance.
118 Stimuli were presented on a 65" CX LG 4K OLED TV (OLED65CXPUA model; LG Electronics Inc.,
119 Seoul, South Korea) with a 60 Hz refresh rate and 3840 x 2160-pixel display resolution. Auditory
120 cues were presented on the display's speaker system to cue the start of the trial and for each
121 change in position of the fixation target. Subjects were placed in a completely dark room and
122 were instructed to complete a perceptual judgment task and provide responses via key presses
123 on a keyboard. The perceptual judgment task was developed in MATLAB™ (R2021b,
124 MathWorks, Natick, MA, USA), utilizing Psychtoolbox (version 3.0.18; Brainard, 1997) and
125 custom MATLAB code for stimulus generation, experiment design, and data acquisition.

126 2.3 Perceptual Judgment Task

127 The current study consisted of a single perceptual judgment task where subjects had to report
128 which among two short vertical lines (0.1 deg wide by 2.6 deg tall) flashed to the left or to the
129 right of the display was closer to a third central line. This variant of Milner's landmark task was
130 used due to its advantage of removing the motor confound that is readily apparent in the line
131 bisection task, with the exception of reporting the two-alternative forced choice response with
132 the right hand via a key press. This is in contrast with an observer having to bisect a line with
133 their hand using a pencil or other bisecting modalities (Bowers & Heilman, 1980; Schenkenberg
134 et al., 1980). Weak within-subject reliability was observed across sensory modalities when using

the line bisection task (Mitchell et al., 2020), deeming the landmark task a more robust method for assessing perceptual asymmetries. The distance of one of two lateral lines varied between 15 and 25 deg eccentricity ranging from 0.25 to 1 deg increments while the other lateral line remained fixed at 20 deg, resulting in 11 possible displacements. There were two possible tasks for each trial - one with eccentric gaze holding (EGH) and one without (see Fig 1). Subjects were instructed to report their answer via the left or right arrow key after being briefly shown the three vertical lines.

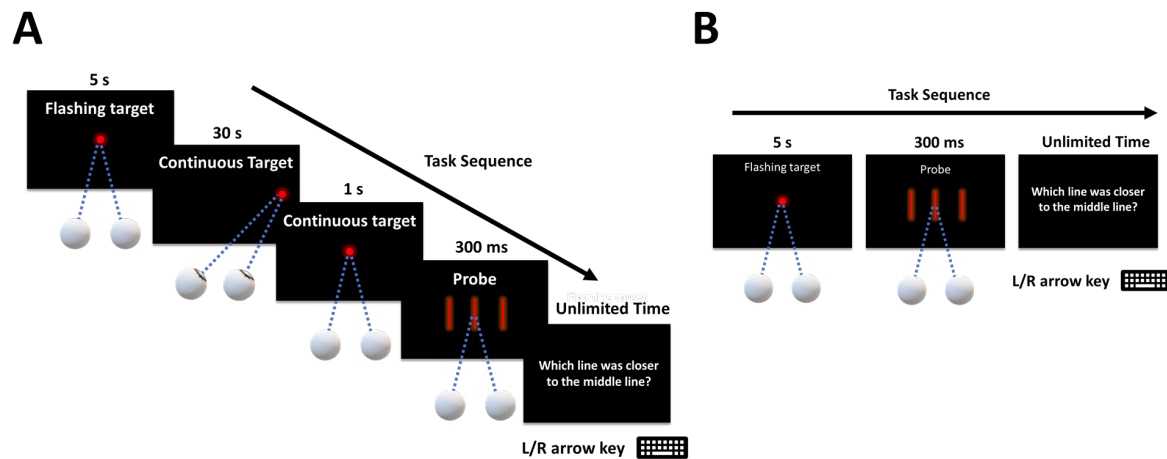


Figure 1. The experimental task sequence with and without eccentric gaze holding. (A) In the experimental condition, subjects had to gaze at an eccentric point for 30 seconds on either the left or right side of the monitor and upon return to central gaze, were presented the three vertical lines. The same judgment task was performed as in the control condition. Subjects were given 1 second to return to central gaze, but the fixation spot was only presented continuously for the initial 500 ms. Also note that subjects were not presented with text at the end of the trial and instead were presented with a black screen. This is to only illustrate that the subject had to

make a choice, whether the left or right line was closer to the central line, at the end of the trial.

(B) In the control condition, subjects were presented with three vertical lines while fixating centrally. Subjects had to report which of two lines were closer to the middle line.

2.4 Experiment Design

Each subject completed two sessions of the perceptual judgment task. The first session consisted of three runs of 176 trials of only the control trials (see Fig 2A) across 11 distance conditions of the lateral lines. The line distances varied between runs in order to establish a set of line distances that would be sufficiently difficult (but not too difficult) for the subject to perform around their threshold performance. After the completion of the three runs, we evaluated the psychometric curves for each run to determine the appropriate line-distance set that was used in the main experiment for the second session. The data from this session were not included in any of the results that are reported in this paper.

The second session required subjects to complete a randomized trial session of 352 trials with and without EGH; gaze holding trials were split into two possible gaze locations – leftward EGH and rightward, both at 40 deg eccentricity. Subjects were tested on 11-line distance conditions and on two gaze holding conditions (with or without EGH). Trials were grouped in 11-trial blocks totaling 32 blocks, with each block reflecting a specific gaze holding condition and gaze location (leftward or rightward gaze at 40 deg), and each trial presenting one of 11 possible line distance conditions for the bisection probe in a randomized order. The EGH trial blocks were grouped in a manner such that the initial trial would require gaze holding for 30 seconds and

the subsequent 10 trials requiring 5 seconds. This trial grouping helped maintain the adaptation making the data collection more efficient and less demanding on the subjects. Trials without EGH were blocked in the same manner but without the gaze holding procedure described previously. Block order was randomized for the entire second session.

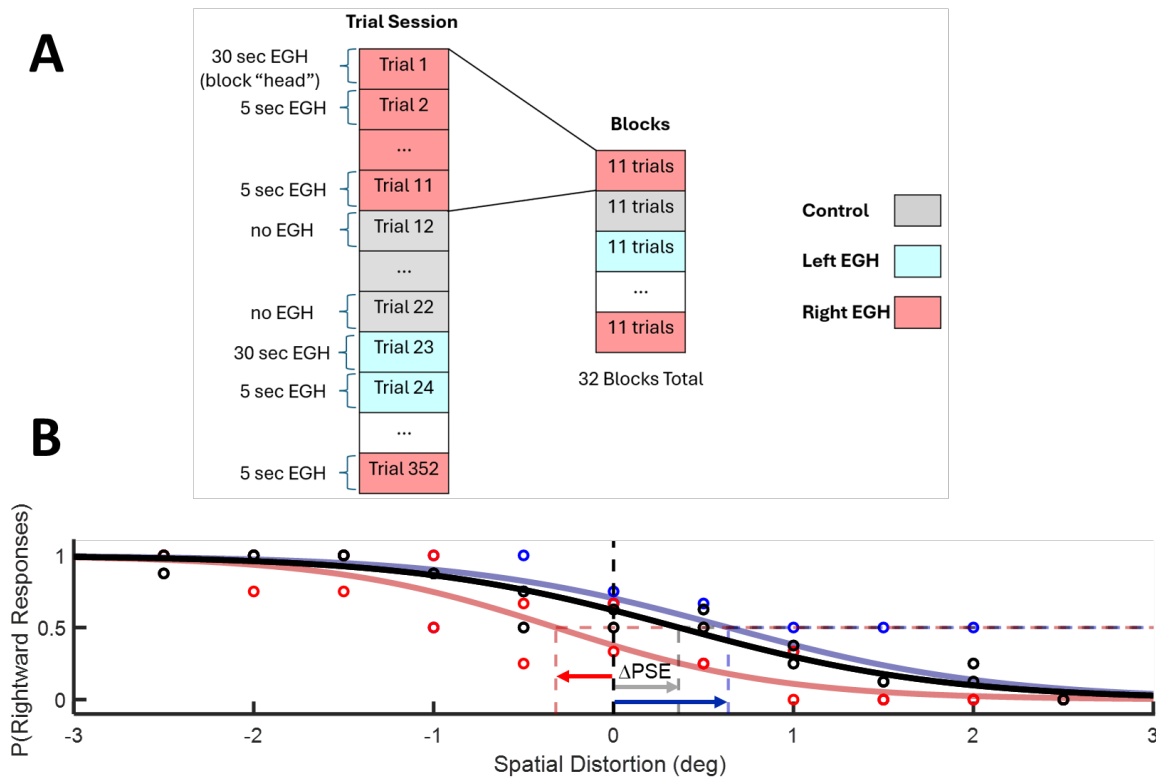


Figure 2. (A) Experimental block design for the eccentric gaze holding experiment. A trial session is divided into 32 blocks, each containing 11 trials of the same gaze-holding condition reflecting the 11 possible line distances. For the eccentric gaze holding conditions, the "head" of the block or starting trial has a 30-second gaze-holding phase and the proceeding 10 trials a 5-second gaze-holding phase to maintain the effects of eccentric gaze holding while reducing the time spent on trial and alleviating physical fatigue. The 32 blocks are pseudo-randomly ordered

181 *within the trial block. (B) Psychometric curves for a specific subject (S11). The intersection of the*
182 *dashed vertical and horizontal lines is simply the point of subjective equality (PSE) for that*
183 *particular gaze condition (specified by its color).*

184 2.5 Statistical Analysis

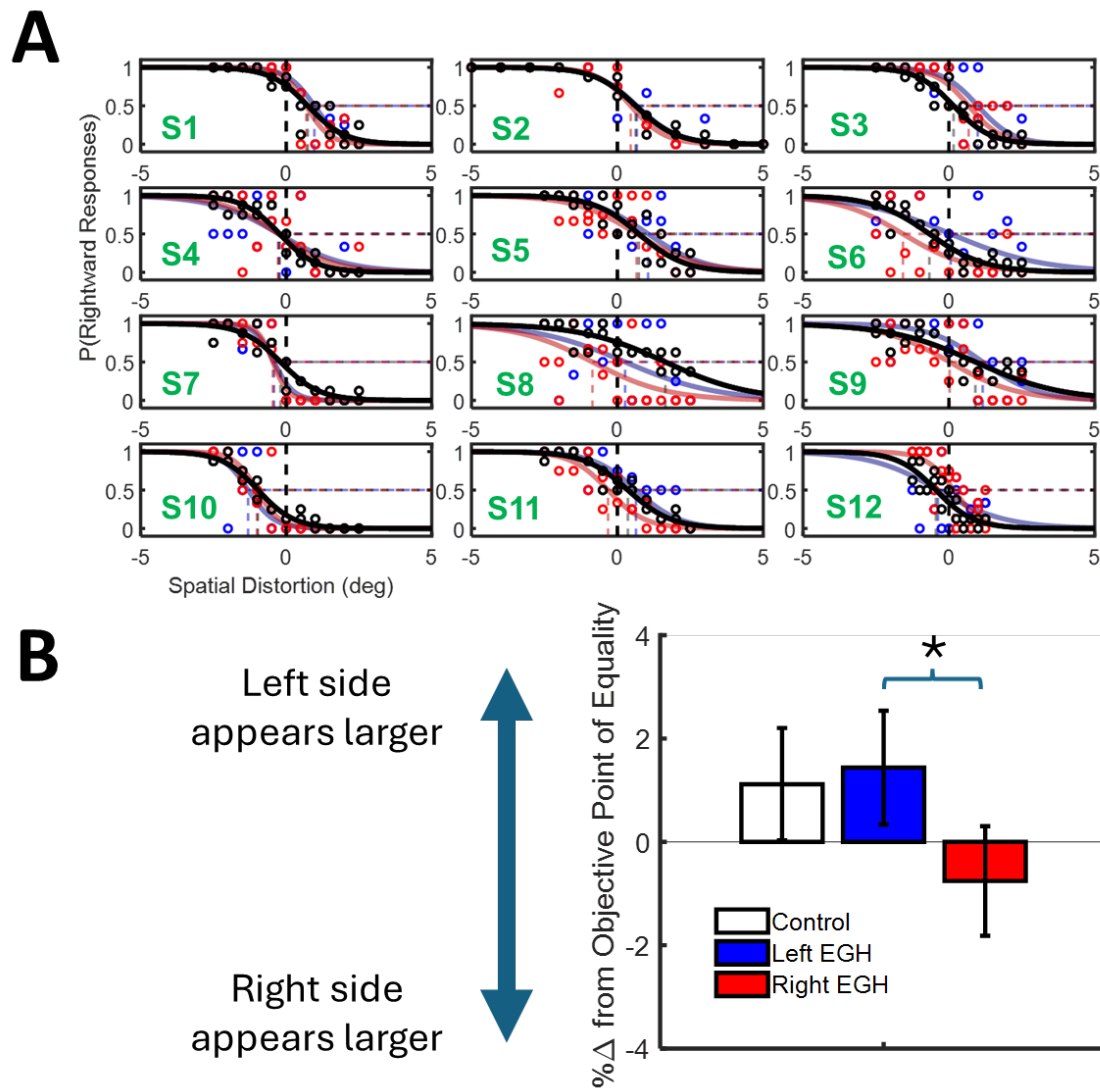
185 All analyses were performed using MATLAB™ (R2021b, R2023a, MathWorks, Natick, MA, USA)
186 and Excel (Microsoft Office Professional Plus 2019, Microsoft Corporation, Redmond, WA, USA).
187 Response data for both gaze holding conditions were each fitted with a generalized linear
188 model (GLM) using a logit link function to model the predicted responses as a binomial
189 distribution. The estimated parameters from the GLM and the line distance condition for a
190 given trial were then used to generate a psychometric function, a sigmoid, for a given
191 experiment block. Two metrics were estimated from the psychometric fits – *spatial bias* and
192 *discrimination sensitivity*. Spatial bias was defined as the point of subjective equality (PSE), in
193 degrees of visual angle, at a proportion of 50% rightward responses. If the PSE is 0 deg, then the
194 subject's perception reflects the perceived relative spacing of the lines, whereas any $|PSE| > 0$
195 would reflect a spatial bias or left-right asymmetry in the perceived distance between the
196 vertical lines. Estimates of discrimination sensitivity were defined as the just-noticeable
197 differences (JND; Fechner, 1860), in degrees of visual angle, indicating the degree of steepness
198 of the psychometric curve. This metric reflects how sensitive the subject is to changes in
199 distance between the left or right line from the center one. JNDs were computed
200 nonparametrically using half the interquartile range of the psychometric curve (Bausenhardt et
201 al., 2018; R. D. Luce & Galanter, 1963).

3 Results

3.1 Spatial bias

Psychometric analysis was performed for all twelve subjects across the three gaze conditions, i.e., left and right EGH, and without EGH. Figure 2B shows three sigmoidal curves for one subject, each representing a gaze holding condition. Curve fits for each subject are shown in Figure 3A.

Since half of the trials for a given EGH condition have the left line adjusted while the right line remains fixed at 20 deg and vice-versa for the other half, the PSE can be visualized in a combined plot by relative changes in visual angle. Subjects, overall, showed average spatial biases of 0.2 ± 0.8 deg (mean $\pm$ SD), 0.3 ± 0.8 deg, and -0.2 ± 0.7 deg for the no EGH (control), left EGH, and right EGH conditions, respectively. For our primary finding, we found that the spatial bias was significantly different between the left and right EGH conditions with a moderate effect size ($t(11) = 2.36$, $p = 0.038$, $d = 0.63$; see Fig 3B), but were not significantly different from the condition without EGH ($p \geq 0.15$). Further, spatial bias under no EGH did not show a significant difference from the point of objective equality (i.e., equidistant lines), although descriptively, showed a slight leftward spatial bias ($p = 0.33$; see Fig 3B).



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219 Figure 3. (A) Psychometric curves of each subject on the relative spatial judgment task.
 220 Scatterplots reflect the proportion of rightward responses for a given line distance condition,
 221 spanning 11 conditions for each experiment. The S-shaped curves reflect the best-fit sigmoid for
 222 a specific experiment. The color scheme indicates the experimental condition for eccentric gaze
 223 holding, where blue and red represent the left and rightward gaze holding conditions,
 224 respectively; the black color scheme is the control experiment without eccentric gaze holding.
 225 (B) Bar plot of the across-subject mean ($\pm$ SEM) of the point of subject equality for twelve

226 *subjects. Units are in percent change from the objective point of equality (when the lateral lines*
227 *are equidistant to the middle line).*

228 *3.2 Secondary findings*

229 The mean ($\pm$ SD) just-noticeable difference for the three gaze holding conditions were 0.9 ± 0.3
230 deg, 1.0 ± 0.5 deg, and 0.8 ± 0.4 deg for the control, left EGH, and right EGH conditions,
231 respectively. Discrimination sensitivity between the three gaze conditions were not significantly
232 different, with the minor exception of left versus right gaze holding conditions revealing
233 marginal significance with a moderate effect size ($t(11) = 2.11, p = 0.06, d = 0.57$), suggesting
234 that prolonged gaze at extreme horizontal eccentricities may have some minor effect on the
235 sensitivity of perceived changes in distance between objects in space between the two
236 eccentric gaze holding conditions. A significant correlation was found between spatial bias and
237 handedness under no EGH ($r^2 = 0.37, p = 0.035$); however, no significant correlations between
238 handedness and the EGH conditions were found ($p > 0.56$). Lastly, a marginally significant
239 correlation was found between discrimination sensitivity and handedness without EGH ($r^2 =$
240 $0.31, p = 0.058$).

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4 Discussion

Under the no gaze holding condition, we expected that the subjects would show a leftward spatial bias where the left side of their visual field would appear larger than the right when the three lines are *objectively* equidistant. Another view of this leftward spatial bias would be when the middle line is objectively closer to the left line when subjects *perceive* the three lines as equidistant. This phenomenon has been reported in many studies under a similar viewing condition while using a similar task paradigm (i.e., the landmark task; Milner et al., 1993). With prolonged gaze holding at an extreme eccentricity, we expected a spatial bias on the side of where gaze was held (see Fig 4). Our data trended in that direction but did not show statistical significance.

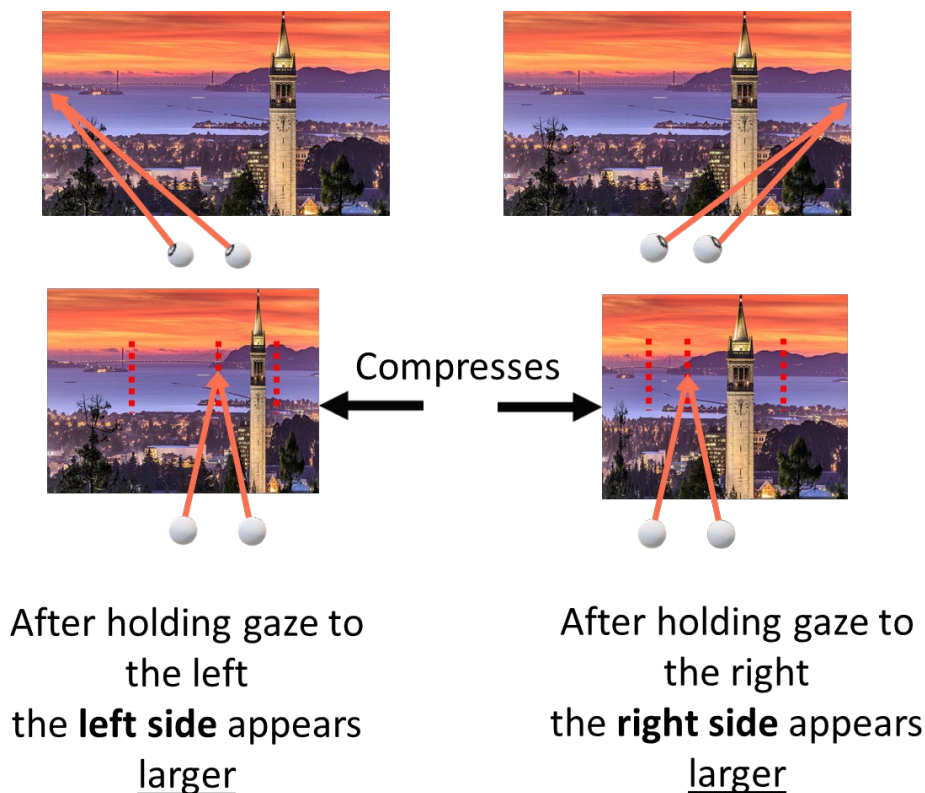


Figure 4. Here two distinct scenarios show a spatial distortion induced from eccentric gaze holding to a particular side. Space becomes compressed on the opposite side of the previously held gaze location, inducing a spatial bias.

Prolonged holding of gaze at an extreme eccentricity invokes an adaptive change in the oculomotor system. Specifically, the oculomotor integrator seems to adapt the set point to the current gaze location. Perceptually, we found that observers showed a spatial bias in the direction of gaze holding upon returning to central fixation, which in turn suggests that observers perceived the left hemifield of visual space as larger than the right after holding gaze to the left and vice-versa. This finding is supported by the significant difference found between the left and right eccentric gaze holding conditions. However, the gaze holding conditions were not statistically significant from the no gaze holding condition. Additionally, differences in the sensitivity to discriminate a range of line displacements were not found across the three gaze holding conditions. This is not surprising since we did not expect gaze holding to necessarily improve or weaken spatial sensitivity upon returning to central gaze. We also did not expect to observe any enhancement or deficits in sensitivity due to directional eye motion (i.e., rebound nystagmus) evoked by prolonged gaze holding since the stimulus probe was brief and did not require the fine visual acuity needed to bolster the spatio-spectral whitening effects of eye motion (Clark et al., 2022; but largely of the random-walk kind).

We found a correlation between handedness and spatial bias under the no gaze holding condition, an association found in a number of previous studies that investigated the pseudoneglect effect (Brodie & Pettigrew, 1996; De Agostini et al., 1999). In contrast, the

correlation dissipates with gaze holding, which could suggest that prolonged gaze holding may play a role in cortical processing of visual space. However, the correlation itself is limited in its interpretability due to the small sample size and the larger proportion of right-handers, which may skew the correlation.

There exists a large corpus of research that shows this perceptual asymmetry or spatial bias. Biases along the visual horizontal meridian were shown with asymmetric retinal stimulation, where the perceived medial plane of the head is biased toward the center of visual field stimulation (Bruell & Albee, 1955). Another bias was found when a frame was presented within a full-field retinal stimulus, where the perceived straight-ahead was shifted toward the center of the frame and not toward the center of retinal stimulation (Broscole, 1967). The current study presents a vertical line in both hemifields and the central vertical line at the center, approximating symmetric retinal stimulation. However, the left and right vertical lines are usually placed at different eccentricities, likely stimulating a different number of receptive fields between both sides of the visual field. This can potentially introduce a bias due to the asymmetry in receptive field stimulation.

Spatial bias can also be introduced via rotations of a surface with a symmetric stimulus on the given surface, for example, two parallel lines as shown in the study by Li and colleagues (Li et al., 2001). Here, the observer instructed the experimenter to move a laser target along a meridian that is parallel and in between the two visible lines where they have to find where they believe to be “straight ahead”. Observers reported biases in the direction of rotation - for

instance, if the surface with two parallel vertical lines was pitched away from the observer, then the observer would perceive their eye level slightly above their actual eye level. If the surface with two parallel horizontal lines was rotated around a vertical axis, then the observer would perceive straight ahead to be slightly to the right, replicating the leftward pseudoneglect effect found in a large corpus of studies (Jewell & McCourt, 2000; Learmonth & Papadatou-Pastou, 2022).

Another consideration is how the brain interprets the varying horizontal offsets of the vertical lines. Can such differences be interpreted as a slanted surface that the lines occupy? A study by Backus and colleagues have shown that vertical size ratio (VSR), the ratio of the vertical subtended angle of the stimulus between the two eyes, affects the perception of azimuthal slant of a surface patch (Backus et al., 1999). Indeed, VSR deviates from 1 when the vertical lines in the current study are not equidistant, which may introduce a bias from the perception of a slanted surface. Based on the empirical results of this study, the size of the vertical lines in our current study should have less than a 25% correction factor for perceived azimuthal slant, and is more impacted by eye position (> 70 % correction factor).

Cortical structures have been investigated extensively to unveil the role of these perceptual asymmetries in perceived visual space. The pseudoneglect phenomena in particular have been linked to a possible hemispheric dominance in parietal cortex for visual attentional processes. A structural imaging study has shown a larger volume of the fronto-parietal tract in the right hemisphere for observers with more pronounced pseudoneglect (de Schotten et al., 2011). Our findings could reflect a possible link between visuo-cognitive processing in higher-order cortex

and low-level adaptive processing in the brainstem and cerebellum. Specifically, gaze holding to a specific side in space could invoke adaptation in multiple parts of the brain, where the brainstem and cerebellum circuits adapt the oculomotor set point for the new gaze location and the fronto-parietal attentional network adapt the perception of visual space - in this case introducing a perceptual bias in the direction of where gaze is being held as suggested by our results. However, further work is needed to truly establish the coordinated adaptation between these two seemingly disparate neural systems.

The eccentricity or spacing of the vertical lines in the current study could also play a role in producing the perception of spatial asymmetry. Previous studies have shown that the length of the line in a line bisection task determines whether the observer would exhibit a pseudoneglect effect or not (Rueckert et al., 2002; Thomas et al., 2012). Specifically, pseudoneglect is more likely to be observed with longer lines than shorter ones. Further, differences in neural activity between the two hemispheres were shown for longer lines (15.3 deg) whereas the shorter lines (1 deg) did not exhibit this difference (Benwell et al., 2014), suggesting that the asymmetry is only present when the stimulus spans a sufficient portion of the visual field. Although, whether this generalizes to horizontal offsets of vertical lines has yet to be seen. Considering that this assumption holds, the eccentricities tested in the current study largely exceed 15 degrees, which in theory should be sufficient to invoke the neural correlates of pseudoneglect.

Another aspect that must be considered is the effect on the initial processing from a specific side or hemifield of visual space. How does the side in which gaze holding takes place play a role in spatial bias? Visual scanning that is initiated from a certain side can elicit a bias on that

side (Brodie & Pettigrew, 1996). Our current paradigm could exacerbate the effects found in these studies that associated the starting hemifield of visual scanning.

Prolonged gaze holding at an eccentric position has been shown to have certain effects on perception and motor control. A study by Harris and Smith (2008) found that eye position in the dark was more accurate after prolonged gaze holding at 11 deg to the left versus holding gaze to the right at 11 deg, revealing a left-right asymmetry. Our findings, although marginal in significance, appeared to reflect this left-right asymmetry in the estimated sensitivity of distinguishing relative positions in space. Qualitatively, we found the spatial bias to be more similar between the left EGH condition and control (mean $\pm$ SEM within-subject difference; left EGH - control: $0.1 \text{ deg} \pm 0.2$, right EGH - control: $-0.4 \pm 0.2 \text{ deg}$), suggesting that perception may deviate from baseline with prolonged gaze holding to the right while also producing a deleterious effect on the accuracy of eye position. However, it is unclear whether prolonged gaze holding harms perceptual performance directly, and consequently, eye-position accuracy, or that the less accurate motor command for eye position impacts perception. This supposed asymmetry in our findings is only speculative and will require further investigation or greater statistical power to reveal a potential veridical effect.

Without gaze holding, a spatial bias exists where the left hemifield appears larger than the right. With eccentric gaze holding, we showed qualitatively that the side that appeared larger was linked to the side that gaze holding took place; however, this effect was not significantly different from the condition without gaze holding. Despite the lack of significance, we showed that spatial bias was different between the left and right EGH conditions, suggesting that

prolonged fixation at extreme eccentricities (40 deg) reveal a differential bias in the perception of space. Although gaze holding invoked differences in the perception of relative distance, it is still unclear whether this perceptual change is causally linked to the change that occurs in the oculomotor neural integrator.

5 Acknowledgments

This work was supported by the NASA Ames Research Center Graduate Fellowship Program, the Department of Veterans Affairs (VA) Veteran Readiness and Employment (Chapter 31) Program, National Institute of Health (NIH) R00 research project grant, and startup lab funds from the University of California, Berkeley. We would like to extend our gratitude to Tiffany Lau for performing a portion of the data collection. We would also like to thank Drs. Martin Banks, Brent Beutter, Susana Chung, Emily Cooper, Dennis Levi, Austin Roorda, Clifton Schor, and Eugene Switkes for their feedback on an earlier iteration of the work. We would also like to thank the members of the Berkeley Ocular Motor Lab (Reem Almagati, Stephanie Reeves, and Drs. Raul Rodriguez and Roksana Sadeghi) for their invaluable feedback and discussions throughout various stages of the study.

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